

Literature Expansion Audit for TreeMAPPO's ICLR 2027 Related Work

Executive summary

The uploaded TreeMAPPO manuscript already has a sensible related-work taxonomy, but the section is **materially thinner than the current literature surrounding its central novelty claim**. Pages 2–4 organize prior work into trajectory-level RL, process supervision, implicit/shared-prefix trees, explicit tree sampling, off-policy tree supervision, agentic branching, and tool-augmented reasoning. The bibliography contains **14 unique references**, and all 14 are effectively consumed by those themes; six of them are concentrated in the comparatively peripheral tool-use/benchmark paragraph, while the rapidly developing fine-grained credit-assignment literature receives much less coverage. [filecite turn0file0](#)

The most important issue is not simply citation count. It is **novelty-neighborhood coverage**. In particular:

Segment Policy Optimization (SPO, NeurIPS 2025) is a conspicuous omission. It explicitly studies segment-level credit assignment, includes a tree-based variant, and contrasts token-, segment-, and trajectory-level advantage estimation. [1](#)

TreeRL (ACL 2025) is even more urgent bibliographically: the manuscript already implements and reports “TreeRL-style branching,” “TreeRL-style credit,” and a combined TreeRL ablation, yet TreeRL has no corresponding entry in the references. TreeRL performs on-policy tree search, branches at high-uncertainty intermediate decisions, and derives dense process supervision from the resulting tree without a separate reward model. [filecite turn0file0](#) [2](#)

The existing **process-supervision paragraph should also be softened**. Its current contrast suggests that denser intermediate supervision fundamentally changes the supervision regime by requiring annotated process labels, verifiers, or reward-model infrastructure. Math-Shepherd automatically constructs process-wise supervision rather than relying on manual step labels, while PRIME goes further and learns implicit process rewards online using policy rollouts and outcome labels. Thus, “outcome-only” by itself is no longer a sufficiently sharp discriminator for TreeMAPPO. [3](#)

The defensible novelty position after incorporating current literature is more specific:

TreeMAPPO combines actively induced, model-plausible token-level counterfactual branching with joint probabilistic inference of latent signed segment contributions and uncertainty from mixed-success descendants, then maps those inferred contributions to token-level policy advantages.

That is considerably safer than implying novelty in token-level branching, tree-structured exploration, or fine-grained critic-free credit assignment individually, because TreeRL, SPO, TreeAdv, GRPO- λ , OAR, PRIME, and OPPO now occupy parts of those spaces. ⁴

I recommend adding **eight papers as the minimum claim-defense set**—SPO, TreeRL, PRIME, GiGPO, GRPO- λ , TreeAdv, OPPO, and Math-Shepherd—and adding OAR, PURE, and Self-Explore for the strongest version of the section. This takes the bibliography from **14 to approximately 22 references with the essential set, or 25 with all eleven**. Those figures are a practical coverage recommendation, **not an ICLR quota**. ICLR 2027's author guidance gives unlimited bibliography pages, and its reviewer guidance asks reviewers to contextualize submissions in the broader literature; the guidance does **not** specify a minimum reference count. It also explicitly says that authors are not required to compare against arXiv-only work, although citing relevant arXiv work when known is strongly encouraged. ⁵

The recommended insertion sequence is: **TreeRL fix $\rightarrow$ SPO $\rightarrow$ Math-Shepherd/PRIME $\rightarrow$ GRPO- λ /OAR/OPPO $\rightarrow$ TreeAdv $\rightarrow$ GiGPO $\rightarrow$ PURE/Self-Explore**. This ordering addresses the most obvious reviewer-facing omissions before adding broader context.

Existing related-work audit

The present section is structurally good. It already separates **what supervision is available** from **how trajectories are explored** and **at what granularity credit is assigned**. That is exactly the right conceptual space for TreeMAPPO. The problem is that the taxonomy was populated mainly by the first wave of tree-based RL papers and older process/tool papers, while several 2024–2026 works now make the boundary around TreeMAPPO substantially more crowded. ^{filecite turn0file0}

A useful way to characterize the current citation distribution is:

Current block	Existing coverage	Assessment
Trajectory-level RL	PPO; DeepSeekMath/GRPO	Adequate foundation, but no modern critic-free method that directly attacks GRPO's uniform credit
Process supervision	Let's Verify Step by Step	Too narrow; automated and outcome-derived process supervision are missing
Shared-prefix tree methods	TEMPO	Reasonable
Explicit tree sampling / advantages	TreeRPO; TreePO	Under-covered; SPO and TreeRL are major omissions
Off-policy tree supervision	Tree-OPO	Reasonable
Agentic branching	APPO	Under-covered; GiGPO is directly relevant

Current block	Existing coverage	Assessment
Tool reasoning / benchmarks	PAL; PoT; ReAct; Toolformer; MATH; DeepMath	Generous relative to the core credit-assignment literature

This produces an unusual density profile: the final tool/benchmark paragraph alone contains nearly half of the manuscript's unique references, even though the paper's contribution is not a new tool-use method or benchmark. Meanwhile, the paragraph most central to novelty—tree-aware and fine-grained advantage estimation—does not yet cite several of the closest available methods. [filecite turn0file0](#)

There are four substantive gaps.

First, the segment-level credit literature is missing. SPO explicitly frames token-level PPO and trajectory-level GRPO as two problematic extremes and proposes segment-level Monte Carlo advantage estimation in between. Its SPO-tree instantiation is especially close to TreeMAPPO's motivation because it uses tree structure to make segment-level estimation efficient on longer reasoning trajectories. A reviewer familiar with NeurIPS 2025 is quite likely to ask about this comparison. [1](#)

Second, the manuscript understates how much of token-level tree branching already exists. TreeRL strategically branches at uncertain intermediate locations during on-policy generation and backs dense supervision out of the resulting tree; TreeAdv likewise explicitly branches group rollouts at high-entropy decisions and redistributes leaf advantages to internal tree segments. Thus, the sentence-level novelty should not rest on “we can branch inside the reasoning trajectory rather than only at tool boundaries or fixed workflow steps.” [6](#)

This matters directly to the manuscript's current method description. TreeMAPPO's stronger differentiator is the **combination** of its branching mechanism and estimator: it scores candidate branch locations using model-side alternative-token probability together with structural penalties, forces the selected alternative as the first token of the new continuation, and then fits a latent additive probabilistic model in which leaf correctness constrains signed segment contributions and segment uncertainty. That particular combination remains distinct from empirical subtree means, local/global value differences, advantage redistribution, or entropy-only branching. [filecite turn0file0](#)

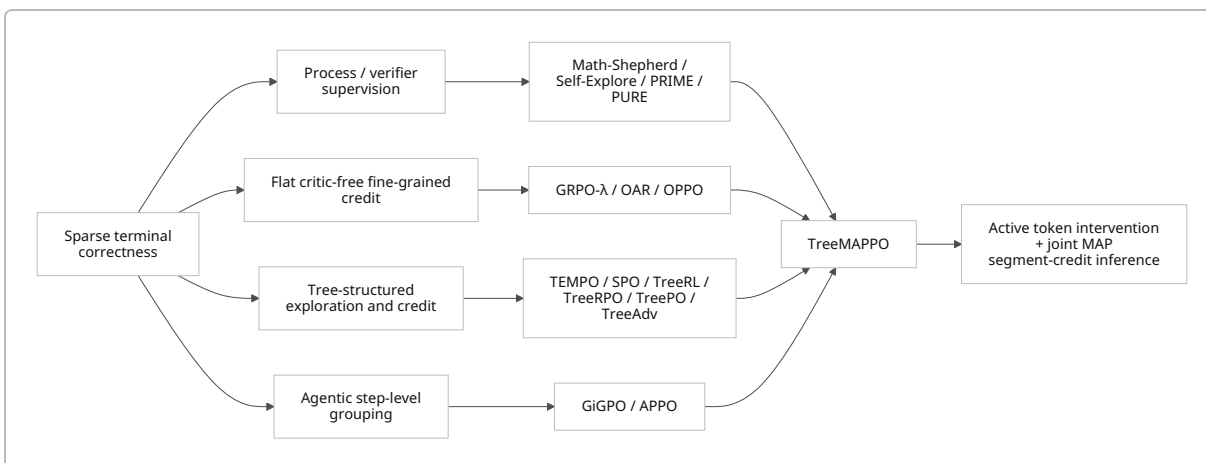
Third, fine-grained critic-free credit no longer implies trees. GRPO- λ derives fine-grained credit using eligibility-trace/ λ -return machinery and a critic-free temporal-difference approximation; OAR reshapes advantages according to counterfactual or gradient-based token influence on the final answer; OPPO obtains a running success probability and token-level advantage through Bayesian recursion over oracle-conditioned evidence. These are especially useful citations because they allow you to state exactly **why the tree is necessary in TreeMAPPO**, rather than leaving flat-rollout alternatives unacknowledged. [7](#)

Fourth, the process-supervision distinction needs modernization. Math-Shepherd automatically constructs stepwise supervision; Self-Explore has models identify the first wrong reasoning step and converts that localization into a fine-grained reward; PRIME trains implicit process rewards online from outcome labels and policy rollouts; PURE demonstrates that even given process rewards, the *credit aggregation rule* can determine whether RL succeeds or reward-hacks. This literature actually strengthens TreeMAPPO's motivation if presented correctly: the clean distinction is **no separately learned process evaluator and no explicit step-level target**, not merely “no human annotations.” [8](#)

A second, concrete editorial problem is the **uncited TreeRL ablation**. Pages 7 and 10–11 describe TreeRL-style branching, TreeRL-style credit, and a combined TreeRL condition, but the bibliography terminates without a TreeRL entry. This should be fixed irrespective of how much additional literature is added.

filecite turn0file0

The literature now suggests the following positioning map:



The diagram's final node is the position I would defend most strongly: **TreeMAPPO is not simply a “tree RL” paper; it is a particular counterfactual-data-generation mechanism paired with a probabilistic latent-credit estimator**. That narrower formulation survives the closest recent literature better. 9

Priority literature and ready-to-insert text

Segment Policy Optimization: Effective Segment-Level Credit Assignment in RL for Large Language Models — Yiran Guo, Lijie Xu, Jie Liu, Dan Ye, Shuang Qiu; 2025; NeurIPS 2025 Main Conference. — Essential

SPO is probably the **single most important missing citation**. It explicitly argues that token-level critic-based advantage estimation and trajectory-level GRPO occupy two undesirable extremes and introduces an intermediate segment-level granularity with flexible segment partitioning and Monte Carlo advantage estimation. Its SPO-tree variant uses tree-based advantage estimation for longer chains of thought and reports improvements over GRPO on MATH500, making it unusually close to TreeMAPPO both motivationally and methodologically. 1

Suggested insertion: “At an intermediate granularity, Segment Policy Optimization (SPO) partitions reasoning traces into flexible segments and estimates segment advantages through Monte Carlo rollouts; its SPO-tree variant exploits tree structure to reduce the sampling cost of advantage estimation (Guo et al., 2025). TreeMAPPO differs by actively intervening at arbitrary token positions and jointly inferring latent signed segment contributions and uncertainty from leaf correctness rather than estimating segment values through Monte Carlo differences.” 1

PDF: [Official NeurIPS PDF](#)

TreeRL: LLM Reinforcement Learning with On-Policy Tree Search — Zhenyu Hou, Ziniu Hu, Yujiang Li, Rui Lu, Jie Tang, Yuxiao Dong; 2025; ACL 2025 Long Papers. — Essential

TreeRL directly integrates on-policy tree search into LLM RL, branches strategically at high-uncertainty intermediate decisions rather than randomly, and obtains dense process supervision from the tree without training a separate reward model. This is indispensable here because the TreeMAPPO manuscript already describes TreeRL-style branching and credit ablations without formally citing the paper. ²

The comparison also forces a useful refinement of TreeMAPPO's novelty. TreeRL already establishes uncertainty-guided intermediate branching; TreeMAPPO should therefore emphasize its **alternative-token intervention policy and joint MAP latent-contribution model**, rather than intermediate branching alone.

²

Suggested insertion: “TreeRL integrates on-policy tree search with tree-derived process supervision, branching preferentially at high-uncertainty intermediate decisions and avoiding a separately trained reward model (Hou et al., 2025). TreeMAPPO instead chooses model-plausible alternative-token interventions using probability and structural criteria and estimates credit through a joint MAP model over signed segment contributions and uncertainty rather than local/global empirical subtree values.” ²

PDF: [ACL Anthology PDF](#)

Process Reinforcement through Implicit Rewards — Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiarui Yuan, Huayu Chen, Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, Ning Ding; 2025; arXiv. — Essential

PRIME is important not because it resembles TreeMAPPO's tree construction, but because it undermines an overly simple “process supervision requires process labels” distinction. PRIME updates an implicit process reward mechanism online using only policy rollouts and outcome labels, producing dense process feedback without expensive human process annotation; it evaluates the approach on competitive mathematics and coding. ¹⁰

TreeMAPPO still has a clear distinction: PRIME constructs an implicit learned process-reward signal, whereas TreeMAPPO attempts to recover localized credit from **experimental outcome variation induced by branched policy continuations**. That is a much sharper contrast.

Suggested insertion: “Recent process-reward methods also reduce or eliminate manual step annotation. PRIME learns implicit process rewards online from policy rollouts and outcome labels, thereby obtaining dense feedback while avoiding a conventional manually annotated PRM pipeline (Cui et al., 2025); TreeMAPPO shares the outcome-only supervision goal but derives local credit directly from branched rollout outcomes rather than learning a separate process-scoring function.” ¹⁰

PDF: [arXiv PDF](#)

Group-in-Group Policy Optimization for LLM Agent Training — Lang Feng, Zhenghai Xue, Tingcong Liu, Bo An; 2025; arXiv. — Essential

GiGPO is a particularly clean addition to the current “Agentic RL and branching granularity” paragraph. It targets the sparse/delayed credit problem in long-horizon LLM agents while retaining critic-free group-based optimization. Its key mechanism is hierarchical relative advantage estimation: full trajectories supply episode-level credit, while repeated “anchor states” across trajectories form local groups for estimating step-level advantages; the paper evaluates ALFWorld and WebShop agents. ¹¹

This gives you an excellent contrast with TreeMAPPO. GiGPO obtains local comparisons when trajectories revisit the same **environment state**; TreeMAPPO deliberately manufactures comparisons by branching a particular **generation prefix/token decision**, making it applicable even when repeated agent states are rare.

Suggested insertion: “For interactive agents, GiGPO obtains critic-free fine-grained credit by combining trajectory-level relative advantages with step-level groups constructed from repeated anchor states across episodes (Feng et al., 2025). TreeMAPPO instead creates local counterfactual comparisons directly at token-level reasoning prefixes, so its credit mechanism does not depend on repeated environment states.” ¹¹

PDF: [arXiv PDF](#)

GRPO- λ : Credit Assignment improves LLM Reasoning — Prasanna Parthasarathi, Mathieu Reymond, Boxing Chen, Yufei Cui, Sarath Chandar; 2025; arXiv. — Essential

GRPO- λ is valuable because it provides a **non-tree counterexample to the idea that structural branching is necessary for finer credit**. It reformulates eligibility traces using token-level log-probabilities and develops a critic-free approximation of the temporal-difference error, effectively bringing λ -return-style temporal credit into GRPO-like reasoning training. The paper studies models from 1.5B to 7B on multiple mathematical reasoning datasets and reports improvements over vanilla GRPO. ¹²

Suggested insertion: “Fine-grained critic-free credit can also be introduced without explicit tree search: GRPO- λ approximates λ -returns and eligibility traces from token-level log-probabilities using a critic-free temporal-difference surrogate (Parthasarathi et al., 2025). This provides a flat-rollout alternative to TreeMAPPO, where localization instead arises from controlled counterfactual branching and posterior inference.” ¹²

PDF: [arXiv PDF](#)

TreeAdv: Tree-Structured Advantage Redistribution for Group-Based RL — Lang Cao, Hui Ruan, Yongqian Li, Peng Chao, Wu Ning, Haonan Song, Renhong Chen, Yitong Li; 2026; arXiv. — Essential

TreeAdv creates an explicit forest out of group-based rollouts, branches at high-entropy/uncertain decisions, and propagates the information in complete-rollout advantages back onto internal tree segments. The authors report evaluation over ten mathematical reasoning benchmarks and position the method as a way to reduce the uniform-credit and length-bias problems of flat GRPO while reducing generated-token cost. ¹³

This paper is close enough that it changes how TreeMAPPO should describe itself. “Tree structure + high-uncertainty branching + internal token/segment advantages” is no longer distinctive; **latent probabilistic contribution inference from patterns of descendant correctness** is.

Suggested insertion: “TreeAdv explicitly branches group rollouts at high-uncertainty decisions and redistributes leaf-level advantages to shared internal tree segments (Cao et al., 2026). In contrast, TreeMAPPO does not directly redistribute or average leaf advantages: it treats segment contributions as latent signed quantities and jointly infers them, together with uncertainty, from the complete pattern of correct and incorrect descendants.” ¹³

PDF: [arXiv PDF](#)

Outcome-Grounded Advantage Reshaping for Fine-Grained Credit Assignment in Mathematical Reasoning — Ziheng Li, Liu Kang, Feng Xiao, Luxi Xing, Qingyi Si, Zhuoran Li, Weikang Gong, Deqing Yang, Yanghua Xiao, Hongcheng Guo; 2026; arXiv. — Strongly recommended

OAR explicitly addresses the same uniform-GRPO-credit weakness but uses attribution rather than rollout trees. OAR-P counterfactually perturbs tokens to estimate their influence on the final-answer distribution; OAR-G approximates that signal with gradients; a conservative bi-level scheme then suppresses low-impact tokens and strengthens pivotal ones while preserving total advantage mass. ¹⁴

This is a good paper for demonstrating that TreeMAPPO is not simply “token importance reweighting.” OAR attributes importance to tokens in an observed trajectory from output sensitivity, whereas TreeMAPPO obtains causal-looking local evidence by **actually sampling alternative continuations** and fitting segment contributions to binary leaf outcomes.

Suggested insertion: “OAR instead attributes credit according to each token's influence on the eventual answer, using either counterfactual token perturbations or an efficient gradient proxy before conservatively reshaping GRPO advantages (Li et al., 2026). TreeMAPPO obtains its local evidence from sampled alternative continuations and fits shared latent segment contributions across the resulting rollout tree rather than attributing an individual completed sequence.” ¹⁴

PDF: [arXiv PDF](#)

OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning — Yu Li, Rui Miao, Tian Lan, Zhengling Qi; 2026; arXiv. — Essential for novelty positioning

OPPO is especially important because both its title and mathematical framing are likely to make reviewers ask how its “Bayesian” credit assignment relates to TreeMAPPO's MAP estimator. OPPO accumulates oracle-conditioned likelihood evidence along a generated trajectory to obtain, in closed form, a running estimate of eventual success probability and corresponding token-level advantages; it does so without a learned value network or additional rollouts, and includes self-oracle and teacher-oracle variants. ¹⁵

The methods are actually quite different, which makes citing it advantageous. OPPO's probabilistic calculation is **sequential value recursion along a trajectory**; TreeMAPPO's is **joint latent-variable**

estimation across a branching incidence structure, with multiple shared segments and mixed-success leaves constraining signed segment means and uncertainty.

Suggested insertion: “OPPO develops a critic-free Bayesian value recursion that converts oracle-conditioned likelihood evidence along a trajectory into running success probabilities and token-level advantages (Li et al., 2026). Although both approaches use probabilistic credit estimation, OPPO performs sequential Bayesian updating along individual trajectories, whereas TreeMAPPO solves a joint MAP problem over latent segment contributions constrained by outcomes across shared branches.” ¹⁵

PDF: [arXiv PDF](#)

Stop Summation: Min-Form Credit Assignment Is All Process Reward Model Needs for Reasoning — Jie Cheng, Gang Xiong, Ruixi Qiao, Lijun Li, Chao Guo, Junle Wang, Yisheng Lv, Fei-Yue Wang; 2025; NeurIPS 2025 Main Conference. — Strongly recommended

PURE studies an adjacent but highly relevant question: **even when fine-grained process rewards exist, how should they be converted into value and advantage signals?** The paper argues that conventional summation-form accumulation encourages exploitation of unusually high process rewards and proposes a min-form value definition based on future process rewards to mitigate reward hacking. It evaluates the method across several base models and reasoning benchmarks. ¹⁶

For TreeMAPPO, PURE is useful evidence that credit assignment is not merely about obtaining dense supervision: the *statistical rule mapping observations to advantages* matters. That directly motivates the manuscript's emphasis on a probabilistically grounded estimator rather than heuristic child-group comparisons.

Suggested insertion: “The aggregation rule for dense rewards can itself be consequential: PURE replaces conventional summation-form process-return credit with a minimum-future-reward value construction to reduce process-reward hacking during reasoning RL (Cheng et al., 2025). TreeMAPPO addresses the complementary outcome-only case, estimating process credit from terminal correctness patterns rather than assuming access to a process reward model.” ¹⁶

PDF: [Official NeurIPS PDF](#)

Self-Explore: Enhancing Mathematical Reasoning in Language Models with Fine-grained Rewards — Hyeonbin Hwang, Doyoung Kim, Seungone Kim, Seonghyeon Ye, Minjoon Seo; 2024; Findings of EMNLP 2024. — Recommended

Self-Explore asks a model to identify the first erroneous step—the “first pit”—inside a reasoning rationale and converts the localized signal into fine-grained rewards. It evaluates on GSM8K and MATH and reports gains over supervised fine-tuning across three LLMs. ¹⁷

It fits naturally immediately after *Let's Verify Step by Step* because it broadens the process-supervision discussion from human-labeled validity to automatically localized error signals. The contrast with TreeMAPPO is straightforward: Self-Explore makes an explicit judgment about where the first error lies;

TreeMAPPO treats local contribution as latent and attempts to infer it from counterfactual branch outcomes.

Suggested insertion: “Self-Explore further reduces reliance on externally labeled process traces by asking the model to locate the first erroneous step in its own rationale and using that ‘first pit’ as a fine-grained reward signal (Hwang et al., 2024). TreeMAPPO does not require explicit error-location judgments; local credit is inferred from outcome differences among descendants of shared trajectory segments.” 17

PDF: [ACL Anthology PDF](#)

Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations — Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, Zhifang Sui; 2024; ACL 2024 Long Papers. — Essential

Math-Shepherd is the clearest correction to the manuscript's current process-supervision wording. It automatically constructs process-wise supervision, trains a reward model that scores individual mathematical reasoning steps, and uses that model both for verification/reranking and reinforcement learning. Its experiments include GSM8K and MATH. 18

The important nuance is that TreeMAPPO can still claim less auxiliary machinery: Math-Shepherd removes **human** process labels but retains an explicit learned process reward model; TreeMAPPO aims to obtain local credit directly from terminal outcome variation in its sampled tree.

Suggested insertion: “Automatic process supervision can also substantially reduce annotation requirements: Math-Shepherd constructs stepwise supervision automatically and trains a process reward model for both verification and reinforcement learning (Wang et al., 2024). TreeMAPPO goes further in removing the learned process-verifier component, deriving local credit directly from correctness variation across policy-generated branches.” 18

PDF: [ACL Anthology PDF](#)

Comparison and integration map

The following table ranks the papers by how directly they bear on TreeMAPPO's claimed contribution. “Essential” below means **essential for a defensible literature/novelty argument**, not an official ICLR requirement. ICLR's own guidance does not impose a numeric bibliography threshold, and arXiv-only papers are explicitly treated more leniently than established peer-reviewed work. 5

Paper	Theme	Main novelty relative to the current draft	Dataset / method	Integration difficulty	Priority
SPO ¹	Segment/tree credit	Flexible segment partition + MC segment advantages; SPO-tree	GSM8K, MATH500; chain/tree MC estimation	Very low: insert into TreeRPO/ TreePO paragraph	Essential
TreeRL ²	On-policy tree RL	High-uncertainty branching + tree-derived process supervision	Math/code; entropy/ uncertainty-guided tree search	Very low; paper already has TreeRL ablations	Essential
PRIME ¹⁰	Outcome-derived process reward	Implicit dense process rewards from rollouts + outcome labels	Competitive math/code; online implicit PRM	Low: process-supervision paragraph	Essential
GiGPO ¹¹	Agentic credit	Episode + anchor-state step-level relative advantages	ALFWorld, WebShop; hierarchical critic-free grouping	Very low: add beside APPO	Essential
GRPO-λ ¹²	Flat critic-free temporal credit	Eligibility-trace/ λ -return credit without a critic	Four math training sets; AIME/MATH/ Olympiad-style evaluation	Low; best in new non-tree paragraph	Essential
TreeAdv ¹³	Tree advantage redistribution	Entropy branching + redistribution of leaf advantages to internal segments	Ten math benchmarks; tree/forest group RL	Very low: direct neighbor to TreeRL/ TreeRPO	Essential
OAR ¹⁴	Token attribution	Perturbation- or gradient-based final-answer influence	Mathematical reasoning; bi-level advantage reshaping	Medium: needs attribution-vs-branching distinction	Strong

Paper	Theme	Main novelty relative to the current draft	Dataset / method	Integration difficulty	Priority
OPPO ¹⁵	Bayesian token credit	Closed-form Bayesian success-value recursion	Seven math/science/code benchmarks; self/teacher oracle	Low: one comparison sentence is enough	Essential for claim defense
PURE ¹⁶	Process reward aggregation	Min-form instead of cumulative process credit	AMC23 + multiple reasoning benchmarks; PRM RL	Low	Strong
Self-Explore ¹⁷	Automated error localization	First-wrong-step self-localization as fine-grained reward	GSM8K, MATH	Very low	Supporting
Math-Shepherd ¹⁸	Automated process supervision	Automatically built process labels + PRM	GSM8K, MATH; verification and RL	Very low	Essential

The key novelty collisions are worth making explicit.

Token-level branching is no longer enough as a novelty claim. TreeRL branches at strategically selected high-uncertainty intermediate locations, and TreeAdv likewise exposes high-uncertainty internal decisions through explicit tree generation. ⁶

Segment-level credit is no longer enough either. SPO is explicitly framed as segment-level policy optimization, while TreeRL and TreeAdv generate intermediate credit from their tree structures. ¹⁹

Critic-free fine-grained credit is also crowded. GRPO- λ , OAR, and OPPO obtain fine-grained signals without learning a standard value critic; GiGPO obtains local agent-action credit while remaining critic-free. ²⁰

Outcome-only supervision is not sufficient as the distinguishing claim. PRIME derives implicit process rewards using policy rollouts and outcome labels, although it still instantiates a learned reward mechanism rather than solving TreeMAPPO's branch-conditioned latent contribution model. ¹⁰

The strongest remaining technical differentiation is therefore the **estimator-plus-intervention combination**. TreeMAPPO actively chooses an alternative token at a model-plausible branch point and forces that counterfactual continuation, then treats the observed leaf outcomes as constraints on shared latent segment contributions m_i and uncertainties u_i under a probit/MAP model. The training signal $m_i / \exp(u_i)$ consequently encodes both direction and inferred certainty. That conceptual centerpiece is

visible in the manuscript's method equations and Figures 1–2 and should become the organizing distinction in Related Work as well. [filecite turn0file0](#)

A compact taxonomy that would make this clear to reviewers is:

Family	How alternatives arise	How local credit is computed	Extra learned evaluator?
GRPO	Independent complete rollouts	One group-normalized outcome per trajectory	No
PRIME	Independent policy rollouts	Learned implicit process rewards	Yes, implicit reward mechanism
GRPO- λ	Flat trajectories	Critic-free temporal/ λ -credit	No conventional critic
OAR	Perturbed/scored trajectory tokens	Outcome-sensitivity attribution	No standard value critic
OPPO	Single trajectories + oracle scoring	Bayesian success-value recursion	Self or frozen teacher oracle
GiGPO	Repeated agent states across episodes	Macro + anchor-state micro relative advantages	No
SPO	Segment partitions / tree MC	Segment Monte Carlo value difference	No critic
TreeRL	Uncertainty-selected tree branches	Tree-derived process/local-global values	No separate reward model
TreeAdv	Entropy-driven tree/forest branches	Redistributed leaf advantages	No
TreeMAPPO	Selected alternative tokens are actively forced into new continuations	Joint MAP inference of latent signed segment contribution + uncertainty	No separate process model

This table is precisely the conceptual comparison that the current Related Work is missing. The TreeMAPPO paper already separates branching-policy ablations from credit-estimator ablations experimentally, so organizing related work along the same two axes would also make the prose agree with the experimental design. [filecite turn0file0](#)

Recommended insertion order and revised paragraph architecture

The least disruptive revision is **not** to append eleven disconnected citations. Instead, preserve the existing subsection and reorganize its middle into a denser progression.

First, repair TreeRL immediately. Add Hou et al. (2025) to the explicit-tree paragraph and to the bibliography. Because TreeRL already appears as a named experimental ablation, omitting the formal citation creates a more serious issue than ordinary literature incompleteness. 2

Second, put SPO next to TreeRPO/TreePO. A natural sequence is TEMPO $\rightarrow$ SPO $\rightarrow$ TreeRL $\rightarrow$ TreeRPO/TreePO $\rightarrow$ TreeAdv. This moves from passive shared-prefix structure, through flexible segment/tree estimation and uncertainty-driven tree search, to explicit tree-relative/group-based credit. 19

A compact replacement for the current explicit-tree material could read:

“Recent methods increasingly exploit structured rollouts to obtain intermediate credit. TEMPO extracts corrections from naturally shared prefixes, while SPO introduces flexible segment-level Monte Carlo estimation and an SPO-tree variant for long reasoning traces. TreeRL performs on-policy tree search by branching preferentially at uncertain intermediate decisions, and TreeRPO/TreePO similarly use explicit tree structure for local comparisons or subgroup-relative advantages. More recently, TreeAdv redistributes leaf advantages over entropy-driven rollout forests. TreeMAPPO is differentiated less by tree construction alone than by the combination of selected token-level counterfactual interventions and joint MAP inference of signed latent segment contributions and uncertainty.” 19

Third, replace the process-supervision paragraph rather than merely extending it. The present wording around Lightman et al. is becoming historically narrow. A stronger paragraph would say:

“A complementary line of work densifies supervision by evaluating intermediate reasoning. Let’s Verify Step by Step demonstrated the value of human-labeled process supervision, while Math-Shepherd automatically constructs stepwise supervision and Self-Explore localizes a model’s first erroneous reasoning step. PRIME further learns implicit process rewards online using policy rollouts and terminal outcome labels, showing that fine-grained process signals need not require manually annotated steps. PURE studies a complementary issue—how dense process rewards should be aggregated into advantages without inducing reward hacking. TreeMAPPO differs from these approaches in that it does not train or query a separate process evaluator or request explicit error-location labels; it infers local contribution from correctness variation across deliberately branched policy trajectories.” 8

That wording preserves the important distinction while avoiding a claim that PRIME and Math-Shepherd would immediately contradict.

Fourth, introduce one short paragraph titled in prose “Fine-grained credit without explicit trees.” This is the highest-value new paragraph because it anticipates the reviewer question: *Why generate additional branches at all?*

“Tree structure is not the only route to critic-free fine-grained credit. GRPO- λ uses a critic-free approximation to λ -return/eligibility-trace credit, OAR reshapes trajectory advantages according to counterfactual or gradient-estimated token influence on the final answer, and OPPO recursively converts oracle-conditioned likelihood evidence into Bayesian token-level success values. These approaches localize credit within flat or individually scored trajectories,

whereas TreeMAPPO actively generates local counterfactual continuations and jointly infers the contributions of shared segments from their descendant outcomes.” 7

That paragraph materially improves the novelty argument with only three new sentences.

Fifth, extend the agentic paragraph with GiGPO before APPO. GiGPO addresses multi-step agent credit by grouping actions at repeated environment states, while APPO focuses on procedural branching at influential points. This creates a useful escalation from environment-state credit to procedural branching to TreeMAPPO's token-level intervention. 11

The resulting Related Work architecture would be:

1. Trajectory-level RL: PPO, GRPO.
2. Process supervision: Lightman, Math-Shepherd, Self-Explore, PRIME, PURE.
3. Tree/segment structured credit: TEMPO, SPO, TreeRL, TreeRPO, TreePO, TreeAdv.
4. Prefix-aware/off-policy tree training: Tree-OPO.
5. Critic-free fine-grained credit without explicit trees: GRPO- λ , OAR, OPPO.
6. Agentic credit and branching: GiGPO, APPO.
7. Tool-augmented reasoning and benchmarks: retain PAL, Program-of-Thoughts, ReAct, Toolformer, MATH, DeepMath.

The actual rewrite can remain compact because many of these papers need only one contrastive clause. In fact, the final related-work section would be stronger if some of the present generic explanation were replaced rather than simply expanded.

For citation prioritization, I would treat the additions as follows:

Claim-defense essential: SPO, TreeRL, PRIME, GiGPO, GRPO- λ , TreeAdv, OPPO, Math-Shepherd.

Strong completeness additions: OAR and PURE.

Useful historical/context addition: Self-Explore.

Adding the eight claim-defense papers raises the reference set from 14 to roughly **22**. Adding all eleven raises it to roughly **25**. Again, I would not describe 22 or 25 as a formal “typical ICLR count”: ICLR 2027 specifies no such threshold. What matters is that the resulting bibliography would finally contain the nearest methods on **all four competing routes to fine-grained credit**—process reward models, flat critic-free attribution, tree-based credit, and multi-step agent credit—rather than being numerically padded with peripheral work. The official ICLR author guide expressly allows unlimited bibliography pages and asks authors to cite related work; reviewer instructions emphasize contextualization in the broader literature.

5

There is also an important venue-status distinction. **SPO, TreeRL, PURE, Math-Shepherd, and Self-Explore are archival conference papers**, so omissions are harder to defend. The cited versions of PRIME, GiGPO, GRPO- λ , TreeAdv, OAR, and OPPO above are arXiv versions. ICLR 2027's reviewer guidance explicitly states that failure to compare against arXiv-only work cannot itself be a basis for rejection, while still encouraging

authors to cite such work when they know it is relevant. Given the extremely direct overlap of several of these papers with TreeMAPPO, proactive citation is nevertheless the safer scholarly choice. ²¹

BibTeX and PDF sources

The entries below are deliberately concise and suitable for merging into the existing `.bib`; conference metadata comes from the official NeurIPS Proceedings or ACL Anthology where available, and arXiv metadata is used for preprints. ²²

Segment Policy Optimization

```
@inproceedings{guo2025segment,
  title      = {Segment Policy Optimization: Effective Segment-Level Credit
               Assignment in RL for Large Language Models},
  author     = {Guo, Yiran and Xu, Lijie and Liu, Jie and Ye, Dan and Qiu,
               Shuang},
  booktitle  = {Advances in Neural Information Processing Systems},
  volume     = {38},
  year       = {2025},
  doi        = {10.52202/085713-3815},
  url        = {https://papers.nips.cc/paper_files/paper/2025/hash/
               a6536243037d1e32c20de85137d478da-Abstract-Conference.html}
}
```

PDF: https://papers.nips.cc/paper_files/paper/2025/file/a6536243037d1e32c20de85137d478da-Paper-Conference.pdf ¹

TreeRL

```
@inproceedings{hou-etal-2025-treerl,
  title      = {{T}ree{RL}: {LLM} Reinforcement Learning with On-Policy Tree
               Search},
  author     = {Hou, Zhenyu and Hu, Ziniu and Li, Yujiang and Lu, Rui
               and Tang, Jie and Dong, Yuxiao},
  booktitle  = {Proceedings of the 63rd Annual Meeting of the Association
               for Computational Linguistics (Volume 1: Long Papers)},
  year       = {2025},
  pages      = {12355--12369},
  publisher  = {Association for Computational Linguistics},
  doi        = {10.18653/v1/2025.acl-long.604},
  url        = {https://aclanthology.org/2025.acl-long.604/}
}
```

PDF: <https://aclanthology.org/2025.acl-long.604.pdf> ²

PRIME

```
@article{cui2025prime,  
  title = {Process Reinforcement through Implicit Rewards},  
  author = {Cui, Ganqu and Yuan, Lifan and Wang, Zefan and Wang, Hanbin  
    and Li, Wendi and He, Bingxiang and Fan, Yuchen and Yu, Tianyu  
    and Xu, Qixin and Chen, Weize and Yuan, Jiarui and Chen, Huayu  
    and Zhang, Kaiyan and Lv, Xingtai and Wang, Shuo and Yao, Yuan  
    and Han, Xu and Peng, Hao and Cheng, Yu and Liu, Zhiyuan  
    and Sun, Maosong and Zhou, Bowen and Ding, Ning},  
  journal = {arXiv preprint arXiv:2502.01456},  
  year = {2025},  
  eprint = {2502.01456},  
  archivePrefix = {arXiv}  
}
```

PDF: <https://arxiv.org/pdf/2502.01456> ¹⁰

GiGPO

```
@article{feng2025gigpo,  
  title = {Group-in-Group Policy Optimization for LLM Agent Training},  
  author = {Feng, Lang and Xue, Zhenghai and Liu, Tingcong and An, Bo},  
  journal = {arXiv preprint arXiv:2505.10978},  
  year = {2025},  
  eprint = {2505.10978},  
  archivePrefix = {arXiv}  
}
```

PDF: <https://arxiv.org/pdf/2505.10978> ¹¹

GRPO- λ

```
@article{parthasarathi2025grpolambda,  
  title = {{GRPO}- $\lambda$ : Credit Assignment improves LLM Reasoning},  
  author = {Parthasarathi, Prasanna and Reymond, Mathieu and Chen, Boxing  
    and Cui, Yufei and Chandar, Sarath},  
  journal = {arXiv preprint arXiv:2510.00194},  
  year = {2025},  
  eprint = {2510.00194},  
  archivePrefix = {arXiv}  
}
```

PDF: <https://arxiv.org/pdf/2510.00194> ¹²

TreeAdv

```
@article{cao2026treeadv,  
  title = {TreeAdv: Tree-Structured Advantage Redistribution for Group-Based  
RL},  
  author = {Cao, Lang and Ruan, Hui and Li, Yongqian and Chao, Peng  
and {Wu Ning} and Song, Haonan and Chen, Renhong and Li, Yitong},  
  journal = {arXiv preprint arXiv:2601.03703},  
  year = {2026},  
  eprint = {2601.03703},  
  archivePrefix = {arXiv}  
}
```

PDF: <https://arxiv.org/pdf/2601.03703> 13

Outcome-Grounded Advantage Reshaping

```
@article{li2026outcomegrounded,  
  title = {Outcome-Grounded Advantage Reshaping for Fine-Grained Credit  
Assignment in Mathematical Reasoning},  
  author = {Li, Ziheng and Kang, Liu and Xiao, Feng and Xing, Luxi  
and Si, Qingyi and Li, Zhuoran and Gong, Weikang  
and Yang, Deqing and Xiao, Yanghua and Guo, Hongcheng},  
  journal = {arXiv preprint arXiv:2601.07408},  
  year = {2026},  
  eprint = {2601.07408},  
  archivePrefix = {arXiv}  
}
```

PDF: <https://arxiv.org/pdf/2601.07408> 14

OPPO

```
@article{li2026oppo,  
  title = {{OPPO}: Bayesian Value Recursion for Token-Level Credit  
Assignment in LLM Reasoning},  
  author = {Li, Yu and Miao, Rui and Lan, Tian and Qi, Zhengling},  
  journal = {arXiv preprint arXiv:2605.21851},  
  year = {2026},  
  eprint = {2605.21851},  
  archivePrefix = {arXiv}  
}
```

PDF: <https://arxiv.org/pdf/2605.21851> 15

PURE

```
@inproceedings{cheng2025pure,  
  title      = {Stop Summation: Min-Form Credit Assignment Is All Process  
               Reward Model Needs for Reasoning},  
  author     = {Cheng, Jie and Xiong, Gang and Qiao, Ruixi and Li, Lijun  
               and Guo, Chao and Wang, Junle and Lv, Yisheng and Wang, Fei-Yue},  
  booktitle  = {Advances in Neural Information Processing Systems},  
  volume     = {38},  
  year       = {2025},  
  doi        = {10.52202/085713-4382},  
  url        = {https://papers.nips.cc/paper_files/paper/2025/hash/  
               be91eb86eb74efc055cff83e953f86ce-Abstract-Conference.html}  
}
```

PDF: https://papers.nips.cc/paper_files/paper/2025/file/be91eb86eb74efc055cff83e953f86ce-Paper-Conference.pdf 16

Self-Explore

```
@inproceedings{hwang-etal-2024-self,  
  title      = {Self-Explore: Enhancing Mathematical Reasoning in Language  
               Models with Fine-grained Rewards},  
  author     = {Hwang, Hyeonbin and Kim, Doyoung and Kim, Seungone  
               and Ye, Seonghyeon and Seo, Minjoon},  
  booktitle  = {Findings of the Association for Computational Linguistics:  
               EMNLP 2024},  
  year       = {2024},  
  pages      = {1444--1466},  
  publisher  = {Association for Computational Linguistics},  
  doi        = {10.18653/v1/2024.findings-emnlp.78},  
  url        = {https://aclanthology.org/2024.findings-emnlp.78/}  
}
```

PDF: <https://aclanthology.org/2024.findings-emnlp.78.pdf> 17

Math-Shepherd

```
@inproceedings{wang-etal-2024-math,  
  title      = {Math-Shepherd: Verify and Reinforce {LLM}s Step-by-step  
               without Human Annotations},  
  author     = {Wang, Peiyi and Li, Lei and Shao, Zhihong and Xu, Runxin  
               and Dai, Damai and Li, Yifei and Chen, Deli and Wu, Yu  
               and Sui, Zhifang},
```

```

booktitle = {Proceedings of the 62nd Annual Meeting of the Association
             for Computational Linguistics (Volume 1: Long Papers)},
year       = {2024},
pages      = {9426--9439},
publisher  = {Association for Computational Linguistics},
doi        = {10.18653/v1/2024.acl-long.510},
url        = {https://aclanthology.org/2024.acl-long.510/}
}

```

PDF: <https://aclanthology.org/2024.acl-long.510.pdf> 18

Taken together, these additions do more than increase citation density. They change the related-work argument from “**TreeMAPPO versus a handful of early tree-RL methods**” into a much stronger statement: multiple recent approaches already provide fine-grained credit through process models, temporal estimators, token attribution, state grouping, segment Monte Carlo estimation, and explicit tree search, while TreeMAPPO occupies the narrower intersection of **actively generated token-level counterfactual branches, outcome-only supervision without a separate process evaluator, and joint uncertainty-aware MAP inference over shared latent segment contributions**. That is the comparison boundary most consistent with the method actually described in the uploaded manuscript and the one least vulnerable to the rapidly expanding 2025–2026 literature. [filecite turn0file0](#) 23

1 9 19 22 23 Segment Policy Optimization: Effective Segment-Level Credit Assignment in RL for Large Language Models

https://papers.nips.cc/paper_files/paper/2025/hash/a6536243037d1e32c20de85137d478da-Abstract-Conference.html?utm_source=chatgpt.com

2 4 6 TreeRL: LLM Reinforcement Learning with On-Policy Tree Search - ACL Anthology

https://aclanthology.org/2025.acl-long.604/?utm_source=chatgpt.com

3 8 18 Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations - ACL Anthology

https://aclanthology.org/2024.acl-long.510/?utm_source=chatgpt.com

5 ICLR 2027 Author Guidelines

https://iclr.cc/Conferences/2027/AuthorGuidelines?utm_source=chatgpt.com

7 12 20 GRPO- λ : Credit Assignment improves LLM Reasoning

https://arxiv.org/abs/2510.00194?utm_source=chatgpt.com

10 Process Reinforcement through Implicit Rewards

https://arxiv.org/abs/2502.01456?utm_source=chatgpt.com

11 Group-in-Group Policy Optimization for LLM Agent Training

https://arxiv.org/abs/2505.10978?utm_source=chatgpt.com

13 TreeAdv: Tree-Structured Advantage Redistribution for Group-Based RL

https://arxiv.org/abs/2601.03703?utm_source=chatgpt.com

14 Outcome-Grounded Advantage Reshaping for Fine-Grained Credit Assignment in Mathematical Reasoning

https://arxiv.org/abs/2601.07408?utm_source=chatgpt.com

15 OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning

https://arxiv.org/abs/2605.21851?utm_source=chatgpt.com

16 Stop Summation: Min-Form Credit Assignment Is All Process Reward Model Needs for Reasoning

https://papers.nips.cc/paper_files/paper/2025/hash/be91eb86eb74efc055cff83e953f86ce-Abstract-Conference.html?utm_source=chatgpt.com

17 Self-Explore: Enhancing Mathematical Reasoning in Language Models with Fine-grained Rewards - ACL Anthology

https://aclanthology.org/2024.findings-emnlp.78/?utm_source=chatgpt.com

21 ICLR 2027 Reviewer Guidelines

https://iclr.cc/Conferences/2027/ReviewerGuidelines?utm_source=chatgpt.com